Advanced Number Theory

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Divisibility rules and prime factorization

Find the prime factorization of 1260.

$1260 = 2^2 \times 3^2 \times 5 \times 7$.

2 [Foundational] • Target: Divisibility rules and prime factorization

Determine whether 143 is prime, and justify your answer.

$143 = 11 \times 13$, so 143 is not prime.

3 [Foundational] • Target: GCF, LCM, and their relationship

Find the GCF and LCM of 36 and 60.

$36 = 2^2 \times 3^2$, $60 = 2^2 \times 3 \times 5$. GCF $= 2^2 \times 3 = 12$. LCM $= 2^2 \times 3^2 \times 5 = 180$.

4 [Foundational] • Target: GCF, LCM, and their relationship

Given that two numbers have GCF 8 and LCM 96, and one of the numbers is 24, find the other number, using the identity GCF × LCM = product of the two numbers.

GCF × LCM $= 8 \times 96 = 768$. Other number $= \frac{768}{24} = 32$.

5 [Foundational] • Target: Modular arithmetic

Find the remainder when 47 is divided by 6, and express this using modular notation.

$47 = 6(7) + 5$, so the remainder is 5. In modular notation: $47 \equiv 5 \pmod 6$.

6 [Foundational] • Target: Modular arithmetic

Find the last digit of 7^{100}, using the pattern of the last digits of powers of 7 (which cycle every 4: 7, 9, 3, 1).

Since the cycle length is 4, find the remainder when 100 is divided by 4: $100 = 4(25) + 0$, so the remainder is 0, which corresponds to the 4th position in the cycle (7,9,3,1): the last digit is 1.

7 [Foundational] • Target: Sums and products of integer sets

Find the sum of all integers from 1 to 100, using the formula $S = \frac{n(n+1)}{2}$.

$S = \frac{100(101)}{2} = 5050$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Sums and products of integer sets

Find the sum of all even integers from 2 to 200.

This is an arithmetic series with first term 2, last term 200, and $n = 100$ terms. Sum $= \frac{n(a_1 + a_n)}{2} = \frac{100(2 + 200)}{2} = \frac{100(202)}{2} = 10100$.

9 [Intermediate] • Target: Number sets and properties

Classify $\sqrt{16}$ as rational or irrational, and explain your reasoning.

$\sqrt{16} = 4$, a whole number, which is rational (it can be written as $\frac{4}{1}$). Even though it involves a square root symbol, the result simplifies to an integer.

10 [Intermediate] • Target: Number sets and properties

Explain why the sum of a rational number and an irrational number is always irrational.

If the sum of a rational number r and an irrational number x were rational (call it s), then $x = s - r$ would be the difference of two rational numbers, which is always rational — contradicting the assumption that x is irrational. So the sum must be irrational.

11 [Intermediate] • Target: Sequences and series

Find the 15th term of the arithmetic sequence with first term 6 and common difference 4.

$a_{15} = a_1 + (15 - 1)d = 6 + 14(4) = 6 + 56 = 62$.

12 [Intermediate] • Target: Sequences and series

Find the sum of the first 8 terms of the geometric sequence $2, 6, 18, 54, \ldots$

First term $a_1 = 2$, ratio $r = 3$. Sum $= \frac{a_1(r^n - 1)}{r - 1} = \frac{2(3^8 - 1)}{2} = 3^8 - 1 = 6561 - 1 = 6560$.

13 [Intermediate] • Target: Absolute value and inequalities with integers

Find all integer solutions of $|x - 5| \le 3$.

$-3 \le x - 5 \le 3$, so $2 \le x \le 8$. Integer solutions: $2, 3, 4, 5, 6, 7, 8$.

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14 **[Intermediate]** • **Target: Combining number theory concepts in a multi-part problem**

This question has three parts. (a) Find the prime factorization of 84 and 126. (b) Use the prime factorizations to find the GCF and LCM. (c) Verify your answer using the identity GCF × LCM = product of the two numbers.

(a) $84 = 2^2 \times 3 \times 7$. $126 = 2 \times 3^2 \times 7$. (b) GCF (using lowest powers of shared primes): $2 \times 3 \times 7 = 42$. LCM (using highest powers of all primes present): $2^2 \times 3^2 \times 7 = 252$. (c) Verify:
GCF × LCM = $42 \times 252 = 10584$. Product of original numbers: $84 \times 126 = 10584$ ✓, confirming the identity holds.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Perfect squares, perfect cubes, and their properties** • **Cross-Lesson: 2.01 Advanced Algebra & Systems**

Find the smallest positive integer n such that $200n$ is a perfect square.

$200 = 2^3 \times 5^2$. For $200n$ to be a perfect square, every prime must appear to an even power. Currently 2^3 needs one more factor of 2 to become 2^4, and 5^2 is already even. So $n = 2$, giving $200(2) = 400 = 20^2$.

16 **[Challenging]** • **Target: Perfect squares, perfect cubes, and their properties** • **Cross-Lesson: 3.01 Advanced Functions**

Determine whether 128 is a perfect cube, and justify your answer using prime factorization.

$128 = 2^7$. For a perfect cube, the exponent must be divisible by 3. Since 7 is not divisible by 3, 128 is not a perfect cube.

17 **[Challenging]** • **Target: Counting divisors of a number** • **Cross-Lesson: 4.01 Advanced Trigonometry**

Find the number of positive divisors of 360, using its prime factorization $360 = 2^3 \times 3^2 \times 5$.

The number of divisors is found by adding 1 to each exponent and multiplying:
$(3 + 1)(2 + 1)(1 + 1) = 4 \times 3 \times 2 = 24$.

18 **[Challenging]** • **Target: Working with consecutive integers and integer constraints** • **Cross-Lesson: 5.01 Pre-Calculus Concepts**

Find three consecutive even integers whose sum is 96.

Let the integers be n, $n + 2$, $n + 4$: $n + (n + 2) + (n + 4) = 96$, so $3n + 6 = 96$, giving $3n = 90$, $n = 30$. The integers are 30, 32, 34.

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19 [Challenging] • Target: Working with consecutive integers and integer constraints • Cross-Lesson: 6.01 Advanced Coordinate Geometry

The product of two consecutive positive integers is 156. Find the integers.

Let the integers be n and $n+1$: $n(n+1)=156$, so $n^2+n-156=0$. Factor: $(n+13)(n-12)=0$, giving $n=12$ (rejecting the negative solution). The integers are 12 and 13.

20 [Challenging] • Target: Digit sum and place value problems • Cross-Lesson: 7.01 Advanced Plane Geometry

A two-digit number has digit sum 11. When the digits are reversed, the new number is 27 more than the original. Find the original number.

Let the original number be $10t+u$ where t is the tens digit and u is the units digit: $t+u=11$. Reversed number: $10u+t$. Given: $(10u+t)-(10t+u)=27$, so $9u-9t=27$, giving $u-t=3$. Adding $t+u=11$ and $u-t=3$: $2u=14$, so $u=7$, then $t=4$. Original number: 47.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Divisibility rules and prime factorization

Find the remainder when 128 is divided by 9.

a. 1

b. 9

c. 2 ✓

d. 3

$128=9(14)+2$, so the remainder is 2.

22 [Foundational] • Target: GCF, LCM, and their relationship

Find the 7th term of the geometric sequence $2, 4, 8, 16, \ldots$

a. 130

b. 125

c. 131

d. 128 ✓

First term $a_1=2$, ratio $r=2$. $a_7=a_1\cdot r^{7-1}=2\cdot 2^6=128$.

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23 [Foundational] • Target: Modular arithmetic

Let a and b be relatively prime positive integers such that $\frac{a}{b} + \frac{14b}{9a}$ is an integer. What is the sum of all distinct possible values of this integer?

a. **8 ✓**

b. 10

c. 16

d. 24

Let $k = \frac{a}{b} + \frac{14b}{9a}$. Combining the fractions over a common denominator gives: $k = \frac{9a^2 + 14b^2}{9ab} \Rightarrow 9abk = 9a^2 + 14b^2$ Since k is an integer, $9ab$ must divide $9a^2 + 14b^2$. We analyze the divisibility constraints: 1. **Divisibility by a:** Since $a \mid 9abk$ and $a \mid 9a^2$, we must have $a \mid 14b^2$. Because $\gcd(a, b) = 1$, $\gcd(a, b^2) = 1$, which implies $a \mid 14$. Thus, $a \in \{1, 2, 7, 14\}$. 2. **Divisibility by b:** Similarly, $b \mid 9abk$ and.

24 [Foundational] • Target: Sums and products of integer sets

A positive integer N has the prime factorization $N = 2^a \cdot 3^b \cdot 5^c$, where a, b, and c are positive integers. If N has exactly 60 positive factors, $2N$ has 72 positive factors, and $3N$ has 80 positive factors, what is the value of c?

a. 2

b. **3 ✓**

c. 4

d. 5

The number of positive factors (divisors) of a number with prime factorization $p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$ is given by $(e_1 + 1)(e_2 + 1) \cdots (e_k + 1)$. 1. For $N = 2^a \cdot 3^b \cdot 5^c$, the number of factors is: $(a + 1)(b + 1)(c + 1) = 60$ 2. Multiplying N by 2 gives $2N = 2^{a+1} \cdot 3^b \cdot 5^c$. The number of factors is: $(a + 2)(b + 1)(c + 1) = 72$ Dividing the second equation by the first equation.

25 [Foundational] • Target: Number sets and properties

The positive integer $N = 2^a \cdot 3^b \cdot 5^c$, where a, b, and c are positive integers, has exactly 60 positive divisors. If $\gcd(N, 600) = 120$ and $\gcd(N, 27) = 9$, what is the value of $a + b + c$?

a. 6

b. 10

c. **12 ✓**

d. 13

First, find the prime factorizations of the given numbers: $600 = 2^3 \cdot 3^1 \cdot 5^2$ $120 = 2^3 \cdot 3^1 \cdot 5^1$ $27 = 3^3$ $9 = 3^2$ Using the property of the greatest common divisor with $\gcd(N, 600) = 120$: - For the prime factor 2: $\min(a, 3) = 3 \Rightarrow a \geq 3$. - For the prime factor 3: $\min(b, 1) = 1 \Rightarrow b \geq 1$. - For the prime factor 5: $\min(c, 2) = 1 \Rightarrow c = 1$ (since c is a positive integer). Next, use $\gcd(N, 27) = 9$: $\min(b, 3) = 2 \Rightarrow b = 2$ The total.

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26 [Foundational] • Target: Sequences and series

How many ordered pairs of positive integers (a, b) satisfy the equation $\gcd(a, b) + \text{lcm}(a, b) = 105$?

a. 20

b. **24** ✓

c. 28

d. 32

Let $g = \gcd(a, b)$. Then we can express $a = gx$ and $b = gy$, where x and y are positive integers such that $\gcd(x, y) = 1$. The least common multiple is $\text{lcm}(a, b) = gxy$. Substituting these into the given equation gives: $g + gxy = 105 \Rightarrow g(1 + xy) = 105$ Since $a, b \geq 1$, we have $x, y \geq 1$, which implies $xy \geq 1$ and $1 + xy \geq 2$. Therefore, g must be a proper positive divisor of $105 = 3 \times 5 \times 7$, so $g < 105$. The possible values of g are.

27 [Foundational] • Target: Absolute value and inequalities with integers

The closed interval $[12, 48]$ is shown on the number line below. How many integers x in this interval satisfy the linear congruence $2x \equiv 1 (\text{mod } 5)$?

a. 6

b. 7

c. **8** ✓

d. 9

Step 1: Solve the linear congruence $2x \equiv 1 (\text{mod } 5)$. Since $\gcd(2, 5) = 1$, the modular inverse of 2 modulo 5 exists. We observe that $2 \times 3 = 6 \equiv 1 (\text{mod } 5)$, so the inverse is 3. Multiplying both sides by 3: $x \equiv 3 (\text{mod } 5)$ Thus, the solutions are integers of the form $x = 5k + 3$ for some integer k. Step 2: Restrict to the closed interval $[12, 48]$: $12 \leq 5k + 3 \leq 48$ Subtract 3 from each part: $9 \leq 5k \leq 45$ Divide by 5.

28 [Foundational] • Target: Combining number theory concepts in a multi-part problem

For a positive integer k, the greatest common divisor of k and 24 is 6, and the least common multiple of k and 40 is 600. What is the value of k?

a. 75

b. 120

c. **150** ✓

d. 300

We use prime factorizations to determine the prime powers dividing k. First, factor the given constants: $24 = 2^3 \cdot 3^1, \quad 40 = 2^3 \cdot 5^1, \quad 600 = 2^3 \cdot 3^1 \cdot 5^2$ From $\text{lcm}(k, 40) = 600$: For each prime p, the exponent in the least common multiple is the maximum of the exponents in k and 40: - For $p = 3$: $\max(v_3(k), 0) = 1 \Rightarrow v_3(k) = 1$. - For $p = 5$: $\max(v_5(k), 1) = 2 \Rightarrow v_5(k) = 2$. - For $p > 5$: $v_p(k) = 0$ since no larger prime divides 600..

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Perfect squares, perfect cubes, and their properties**

What is the smallest positive integer n such that $126n$ is the square of an integer?

a. 7

b. 14 ✓

c. 21

d. 42

To find the smallest positive integer n such that $126n$ is a perfect square, first find the prime factorization of 126: $126 = 2 \times 63 = 2 \times 3^2 \times 7$ For an integer to be a perfect square, every prime factor in its prime factorization must have an even exponent. In the prime factorization of 126: - The factor 2 has an exponent of 1 (odd). - The factor 3 has an exponent of 2 (even). - The factor 7 has an exponent of 1 (odd). To make all exponents even with the smallest positive integer n, n must.

30 **[Intermediate]** • **Target: Counting divisors of a number**

The polynomial $P(x) = x^4 - 8x^3 + ax^2 + bx + c$ has four real roots that form an arithmetic progression. If the sum of the squares of the four roots is 36, what is the value of the constant term c?

a. -15 ✓

b. -9

c. 9

d. 15

Let the four real roots in arithmetic progression be represented symmetrically as $m - 3d$, $m - d$, $m + d$, and $m + 3d$, where $2d$ is the common difference. By Vieta's formulas, the sum of the roots is the negative of the coefficient of x^3: $(m - 3d) + (m - d) + (m + d) + (m + 3d) = 4m = 8 \Rightarrow m = 2.$ Next, the sum of the squares of the roots is given as 36: $(2 - 3d)^2 + (2 - d)^2 + (2 + d)^2 + (2 + 3d)^2 = 36.$ Expanding each term.

31 **[Intermediate]** • **Target: Working with consecutive integers and integer constraints**

What is the remainder when 4^{103} is divided by 11?

a. **9** ✓

b. 5

c. 4

d. 3

We want to find the value of $4^{103} \pmod{11}$. By Fermat's Little Theorem, if p is a prime number and $\gcd(a, p) = 1$, then: $a^{p-1} \equiv 1 \pmod{p}$ Since 11 is prime and $\gcd(4, 11) = 1$, we have: $4^{10} \equiv 1 \pmod{11}$ We express the exponent 103 in terms of multiples of 10: $103 = 10 \times 10 + 3$ Using exponent laws and modular arithmetic: $4^{103} = (4^{10})^{10} \times 4^3 \equiv (1)^{10} \times 4^3 \equiv 4^3 \pmod{11}$ Now calculate 4^3.

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32 [Intermediate] • Target: Digit sum and place value problems

Find the greatest common factor (GCF) of 53 and 55.

a. 3

b. 2

c. 4

d. **1** ✓

$53 = 1 \times 53$ and $55 = 1 \times 55$, and 53, 55 share no common factor, so $\gcd(53, 55) = 1$.

33 [Intermediate] • Target: A multi-part modular arithmetic and remainder problem

An arithmetic sequence has $a_1 = 3$ and $d = 6$. If the sum of the first n terms is 300, find n.

a. 11

b. 9

c. 5

d. 10 ✓

$S_n = \frac{n}{2}(2a_1 + (n-1)d) = 300$, and solving the quadratic gives $n = 10$.

34 [Intermediate] • Target: Divisibility rules and prime factorization

Find the least common multiple (LCM) of 10 and 20.

a. 200

b. 30

c. **20** ✓

d. 10

$\text{lcm}(10, 20) = \frac{10 \times 20}{\gcd(10, 20)} = \frac{200}{10} = 20$.

35 [Intermediate] • Target: GCF, LCM, and their relationship

Find the sum of the first 10 terms of the arithmetic sequence with $a_1 = 3$ and $d = 1$.

a. -75

b. 150

c. $\frac{165}{2}$

d. 75 ✓

$a_{10} = 3 + 9(1) = 12, \quad S_{10} = \frac{10}{2}(3 + 12) = 75$

36 [Intermediate] • Target: Modular arithmetic

An arithmetic sequence has $a_2 = 6$ and $a_7 = -4$. Find a_4.

a. 0

b. 10

c. 2 ✓

d. -2

$d = \frac{a_7 - a_2}{7-2} = \frac{-4-(6)}{5} = -2, \quad a_1 = a_2 - (2-1)d = 8, \quad a_4 = 8 + (4-1)(-2) = 2$

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37 [Intermediate] • Target: Sums and products of integer sets

What is the common difference of the arithmetic sequence $0, -2, -4, -6, \ldots$?

a. 0 **b.** -2 ✓

c. -4 **d.** -3

Applying Sequences & Series principles, $d = -2 - 0 = -2$ Therefore, the verified correct option is -2.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Number sets and properties • Cross-Lesson: 5.01 Pre-Calculus Concepts

An arithmetic sequence has first term $a_1 = -7$ and common difference $d = -1$. Find a_6.

a. -2 **b.** -12 ✓

c. 12 **d.** -13

$a_6 = a_1 + (6-1)d = -7 + 5(-1) = -12$

39 [Challenging] • Target: Sequences and series • Cross-Lesson: 6.01 Advanced Coordinate Geometry

An infinite geometric sequence $a_1, a_2, a_3, \ldots$ of positive real numbers has common ratio $r < 1$ and sum S. The sub-sequence of terms whose indices are of the form $3k-1$ for positive integers k (that is, $a_2, a_5, a_8, \ldots$) has sum $\sum_{}^{\infty} a_{3k-1} = \frac{3}{13}S$. Furthermore, the infinite series formed by the squares of the terms has sum $\sum_{n=1}^{\infty} a_n^2 = \frac{9}{2}S$. What is the value of $\sum_{n=1}^{\infty} na_n$?

a. $\frac{27}{2}$ ✓ **b.** $\frac{81}{4}$

c. $\frac{9}{2}$ **d.** $\frac{27}{4}$

Let the first term be $a_1 = a > 0$ and the common ratio be $r \in (0, 1)$. The sum of the original series is: $S = \sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}$. **Step 1: Determine the common ratio r.** The sub-sequence consists of terms $a_2, a_5, a_8, \ldots$, which has first term $a_2 = ar$ and common ratio r^3. Its sum is: $\sum_{k=1}^{\infty} a_{3k-1} = \frac{ar}{1-r^3}$. Given that this sum is $\frac{3}{13}S$: $\frac{ar}{1-r^3} = \frac{3}{13} \cdot \frac{a}{1-r}$. Since $a > 0$ and.

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40 [Challenging] • Target: Absolute value and inequalities with integers • Cross-Lesson: 7.01 Advanced Plane Geometry

Let $(a_n)_{n \geq 1}$ be an arithmetic sequence of real numbers with first term $a_1 > 0$ and non-zero common difference d. The terms a_1, a_3, and a_4 form the first three terms of a convergent geometric sequence $(g_k)_{k \geq 1}$, such that $g_1 = a_1$, $g_2 = a_3$, and $g_3 = a_4$. Let $S_\infty = \sum_{k=1}^{\infty} g_k$ denote the sum to infinity of the geometric sequence, and let $S_n = \sum_{j=1}^{n} a_j$ denote the n-th partial sum of the arithmetic sequence. If $M = \max_{n \geq 1} S_n$ is the maximum value attained by the partial sums of the arithmetic sequence, what is the value of $\frac{M}{S_\infty}$?

a. $\frac{5}{4}$ ✓

b. $\frac{81}{64}$

c. $\frac{9}{8}$

d. $\frac{5}{8}$

1. **Determine the relation between d and a_1:** The terms of the arithmetic sequence are $a_n = a_1 + (n-1)d$. Therefore: $g_1 = a_1, \quad g_2 = a_3 = a_1 + 2d, \quad g_3 = a_4 = a_1 + 3d$ Since g_1, g_2, g_3 form a geometric progression, the geometric mean property requires: $g_2^2 = g_1 g_3 \Rightarrow (a_1 + 2d)^2 = a_1(a_1 + 3d)$ Expanding both sides: $a_1^2 + 4a_1 d + 4d^2 = a_1^2 + 3a_1 d \; a_1 d + 4d^2 = 0 \Rightarrow d(a_1 + 4d) = 0$ Since $d e 0$, we have $a_1 + 4d = 0$, which gives.

41 [Challenging] • Target: Combining number theory concepts in a multi-part problem • Cross-Lesson: 8.01 Statistics, Counting & Probability

Prove $\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$ using the formula for arithmetic sums.

a. Cannot be proved simply

b. By FTA

c. **By induction: base $n = 1$ trivially true; induction step involves algebraic manipulation** ✓

d. Use telescoping

Applying Sequences & Series principles, Standard induction. Therefore, the verified correct option is By induction: base $n = 1$ trivially true; induction step involves algebraic manipulation.

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42 [Challenging] • Target: Perfect squares, perfect cubes, and their properties • Cross-Lesson: 2.01 Advanced Algebra & Systems

Use generating functions to find the number of ways to write n as an ordered sum of positive integers.

a. n^2

b. 2^{n-1} ✓

c. $n!$

d. 2^n

Applying Sequences & Series principles, Compositions of n: 2^{n-1}. Therefore, the verified correct option is 2^{n-1}.

43 [Challenging] • Target: Counting divisors of a number • Cross-Lesson: 3.01 Advanced Functions

A storage bin contains red, blue, and green blocks. The ratio of red blocks to blue blocks is $3:5$. After 12 green blocks are added to the bin, with no other blocks added or removed, the total number of blocks in the bin is 60, and the probability of randomly selecting a green block from the bin is $\frac{1}{3}$. How many red blocks were in the bin originally?

a. 12

b. **15** ✓

c. 18

d. 25

1. Find the total number of green blocks after the addition: Since the new total is 60 and the probability of selecting a green block is $\frac{1}{3}$, the number of green blocks is: $60 \times \frac{1}{3} = 20$ 2. Determine the number of red and blue blocks: The remaining blocks in the bin are red and blue. Since green blocks account for 20 of the 60 blocks, the combined total of red and blue blocks is: $60 - 20 = 40$ (Alternatively, originally there were $20 - 12 = 8$ green blocks, and the original total was.

44 [Challenging] • Target: Working with consecutive integers and integer constraints • Cross-Lesson: 4.01 Advanced Trigonometry

The bar graph below shows the number of students enrolled in each of four elective courses at a high school. If one student is selected at random from these four courses, what is the probability that the selected student is enrolled in Music?

a. $\frac{7}{50}$

b. $\frac{7}{20}$ ✓

c. $\frac{7}{13}$

d. $\frac{13}{20}$

To find the probability that a randomly chosen student is enrolled in Music, divide the number of students enrolled in Music by the total number of students enrolled across all four courses. 1. Find the total number of students: $\text{Total} = 8 + 14 + 6 + 12 = 40$ 2. Find the number of students in Music: $n(\text{Music}) = 14$ 3. Calculate the probability: $P(\text{Music}) = \frac{14}{40} = \frac{7}{20}$ Common misconceptions: - $\frac{7}{13}$ computes the odds in favor (ratio of students in Music to studen.

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45 [Challenging] • Target: Digit sum and place value problems • Cross-Lesson: 5.01 Pre-Calculus Concepts

What is the sum of all real values of k for which the equation $|x^2 - 4| = 2x + k$ has exactly three distinct real solutions?

a. 4

b. 5

c. 7

d. **9** ✓

We analyze the intersections of the function $f(x) = |x^2 - 4|$ with the family of parallel lines $y = 2x + k$. The function $f(x)$ is defined piecewise: - Outer branches (for $|x| \geq 2$): $y = x^2 - 4$ - Inner branch (for $|x| \leq 2$): $y = 4 - x^2$ We consider intersections on each branch: 1. **Inner branch ($-2 \leq x \leq 2$):** $4 - x^2 = 2x + k \Leftrightarrow x^2 + 2x + (k - 4) = 0$ The discriminant of this quadratic is:

$\Delta_{\text{inner}} = 2^2 - 4(1)(k - 4) = 4 - 4k + 16 = 20 - 4k = 4(5 - k)$ - If.

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